

FreeReg: Free-Coordinate Registration of 3D Models for Augmented Reality in Image-Guided Bronchoscopy

Ranyang Li, Nan Wei, Xiaodong Guo, Chao Fan, Zhipeng Lin, Wufeng Liu, and Quncheng Zhang

Abstract—Image-guided bronchoscopy and transbronchial biopsy depend on registering a preoperative CT airway model onto the live endoscopic view, yet electromagnetic navigation reports 5–10 mm clinical error, loses track under respiration, and requires an external field generator. Classical calibrate-then-align pipelines assume a calibrated camera, a rigid airway, and independent per-frame poses—assumptions that fail under uncalibrated optics, continuous tissue deformation, and temporal jitter. We formulate CT-to-bronchoscopy registration as free-coordinate estimation of the projection mapping among model, camera, and pixel frames, allowing intrinsics, depth, rigid pose, and deformation to be optional and degradable. FreeReg combines a centerline-driven correspondence engine that registers the airway where depth-based ICP collapses, an observability-guided rigid-plus-warp estimator that deforms only within the spectrally observable subspace of the single-view depth-residual operator, a manifold $SE(3)$ Kalman filter with innovation gating, and a co-anchored registration-rendering field whose photometric residual provides online, label-free validation. On 12 clinical CT airways (270 synthetic fly-through frames with ground-truth poses), FreeReg achieves 4.23 mm mean target registration error (92% under 10 mm) at 30–80 FPS, within the accuracy range of electromagnetic navigation without external tracking hardware; deformation recovery reaches 1.1% of model extent on a real-colon benchmark, and real intra-operative video supports qualitative AR demonstration. On rigid benchmarks the engine remains correct (competitive ADD-S on BOP Linemod), yet its differentiating strength is in the airway, where general-purpose pose and reconstruction methods (FoundationPose 104 mm, Endo3R 34 mm) fail out-of-distribution. The observability analysis quantifies which deformation directions monocular endoscopy can and cannot recover, giving explicit clinical limits on image-guided biopsy.

Index Terms—Bronchoscopy, CT-to-endoscopy registration, image-guided intervention, non-rigid registration, surgical navigation, transbronchial biopsy, observability, Kalman filtering, augmented reality, Gaussian splatting.

I. INTRODUCTION

Motivation and significance. Lung cancer remains the leading cause of cancer death, and its definitive diagnosis frequently requires transbronchial biopsy of a peripheral lesion visible only on preoperative CT. During bronchoscopy the clinician steers a camera through a deformable, poorly textured, fluid-covered airway with no external landmarks, so the lesion seen on CT is not directly visible through the scope. The clinical bottleneck is therefore a registration problem: aligning the preoperative CT airway and target onto the live

endoscopic view. Deployed electromagnetic navigation bronchoscopy (ENB) addresses this with an external field generator and tracked probe, but still reports 5–10 mm localization error [?], loses track under respiration and coughing, and adds hardware and setup burden. A camera-only, calibration-free registration that reaches the accuracy and speed required for image-guided biopsy would directly raise diagnostic yield and reduce complications such as pneumothorax and hemorrhage. This is the problem this paper addresses.

The registration problem and its classical solution. At its core CT-to-bronchoscopy alignment is a projection-mapping problem among at least three coordinate frames: the *model frame* $\mathcal{F}_m$ (the CT airway reconstruction and target), the *camera frame* $\mathcal{F}_c$ (the bronchoscope tip), and the *pixel frame* $\mathcal{F}_p$. A point $\mathbf{X}_m$ in the model frame is carried to the image by the chain $\mathbf{x}_p = \pi_D(K \mathbf{T}_{cm}(\mathbf{X}_m + W(\mathbf{X}_m)))$, where $\mathbf{T}_{cm} = (\mathbf{R}, \mathbf{t}) \in SE(3)$ is the rigid model-to-camera pose, W is a deformation field, K collects the camera intrinsics, and π_D is the (possibly distorted) perspective projection. The classical decomposition (calibrate the camera, collect corresponding points between model and scene, and align them by a rigid rotation-translation (RT) transform [?], [?], [?]) rests on three assumptions that fail in the airway. *First, calibration is unavailable or drifts:* intrinsics K are rarely provided for a deployed bronchoscope and change with focus and coupling. *Second, the airway is not rigid:* respiration, cardiac pulsation, and instrument pressure deform it continuously, so a rigid fit reallocates the deformation into pose error and the overlay appears to “swim.” *Third, per-frame estimation is temporally incoherent:* independent per-frame poses jitter, producing floating and flicker that break the guidance illusion and can mislead the operator.

Why general-purpose methods fail in the airway. Object pose estimation has advanced rapidly: on the BOP benchmark, methods progressed from MegaPose to SAM-6D and FoundationPose, which now saturate Linemod [?], [?], [?]. Calibration-free reconstruction (DUST3R, VGGT) and endoscopic reconstruction (Endo3R, EndoNerF) recover geometry from uncalibrated video [?], [?], [?], [?], and neural rendering (3D Gaussian Splatting) delivers real-time photorealistic views [?]. Yet these advances are built for rigid objects, controlled illumination, or offline reconstruction, not for registering a known external model into a live, deforming, calibration-free endoscopic stream. This is precisely why they cannot be dropped into the bronchoscopy suite: we show quantitatively that the champions collapse in the airway, where FoundationPose’s appearance-based scorer fails on surgical imagery (TRE 104 mm), Endo3R drifts (ATE 34 mm), and monocular depth on featureless mucosa is unreliable. The gap is structural, not incremental, and it motivates a geometry-first method designed for the endoscopic setting itself.

Free-coordinate registration and observability. The general task is model-to-image/video AR/MR registration: given a 3D model in its native coordinates and an image or video stream, estimate an explicit mapping into camera and pixel coordinates, then render a temporally stable, depth-ordered model composite. This is coordinate-system registration, not unconstrained visual overlay. Rather than assuming a calibrated world frame, we treat the mapping $F: \mathcal{X}_m \leftrightarrow \mathcal{X}_c \leftrightarrow \mathcal{X}_p$

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as jointly estimable, with intrinsics K , depth z , rigid pose $(\mathbf{R}, \mathbf{t})$, deformation W , and scale each optional and degradable. We call this *free-coordinate registration*, because no frame is privileged as a fixed reference; factors are estimated from available evidence or clamped when unobserved: $F = \pi_D \circ K \circ (\mathbf{R}, \mathbf{t}) \circ (I + W)$. This framing exposes the central limit: *which factors are observable from the data at all?* The single-view depth-residual operator is severely rank-deficient, so most deformation directions cannot be inferred regardless of optimizer. We isolate this structure and use it to constrain estimation to the observable subspace (Section IV-B).

In summary, this work makes the following contributions.

- A **calibration-free CT-to-bronchoscopy registration pipeline** reaching 4.23 mm mean target registration error (92% under 10 mm) at 30–80 FPS on 12 clinical CT airways (ground-truth fly-throughs), within the accuracy range of electromagnetic navigation, via a centerline-driven correspondence engine that works where depth-based ICP collapses (Section V).
- **Observability-guided non-rigid registration** (Section IV-B): the single-view depth-residual operator is rank-deficient to 151/648 directions (23%); we deform only within the observable subspace, giving the first explicit account of which airway deformations monocular endoscopy can and cannot recover.
- A **free-coordinate AR/MR formulation and correspondence engine** (Sections III, IV-A) mapping a native 3D model to camera/pixel coordinates for anchored, occlusion-aware visualization, unifying calibrated/uncalibrated RGB/RGB-D inputs and resolving symmetric ambiguities by silhouette (9/9 versus 5/9 for depth), with a learned front-end at $5\times$ speed.
- **Gated manifold temporal consistency and co-anchored rendering** (Sections IV-C, IV-D): an SE(3) Kalman filter with innovation gating, plus one deformation field driving both registration and a Gaussian-splatting renderer whose photometric residual is an online validation signal.

Figure 1 summarizes what FreeReg does end-to-end. The full pipeline, evaluation scripts, and the Windows build of FoundationPose used for comparison are released for reproducibility.

II. RELATED WORK

Object pose estimation. BOP-benchmark 6-DoF pose estimation has progressed from classical feature methods [?] through MegaPose (AR ≈ 0.53) [?] and SAM-6D (AR 0.818–0.825) [?] to FoundationPose (AR 0.847) [?], the first unified CAD/model-free zero-shot method. These are per-frame, rigid, intrinsics-dependent estimators. Our engine is interface-isomorphic with FoundationPose’s output (a

4×4 pose) and is swappable with it; we additionally compare against its official implementation, ported to Windows for this work, on shared frames. Classical correspondence solvers underpin all of these: Kabsch [?] and Umeyama [?] for 3D–3D alignment, ICP [?], FPFH+RANSAC [?], [?], and EPnP [?] for 2D–3D.

Calibration-free reconstruction. DUST3R (CVPR 2024) [?] regresses pointmaps from uncalibrated image pairs; MAST3R and VGGT [?] extend it to larger feeds. Endo3R (MICCAI 2025) [?] specializes to endoscopic video but yields unreliable intrinsics on our sequences. Monocular SLAM (ORB-SLAM3 [?], DROID-SLAM [?]) provides trajectories without metric scale or external-model registration. These methods recover scene geometry but do not register a known external model into the camera frame, our target task.

Non-rigid registration. Free-form deformation (FFD) [?], [?] and embedded-deformation graphs [?] parameterize smooth warps by a sparse control lattice; DynamicFusion [?] reconstructs non-rigid scenes in real time; NRSfM factors deformation from image correspondences; 4D Gaussian Splatting [?] renders dynamics but optimizes for reconstruction, not registration. In the surgical domain, Mountney *et al.* recover and track 3-D tissue deformation from endoscopic video [?], and Stoyanov *et al.* reconstruct stereo geometry in robot-assisted surgery [?], both relying on stereo or tracking priors. Crucially, *none analyzes the observability of deformation under single-view projection*, which we show is rank-deficient to 23%. Our spectral observability decomposition, and the estimator it motivates, is to our knowledge new in this setting.

Temporal consistency and medical AR. Manifold filtering on SE(3) is standard in robotics (error-state EKF) [?], [?]. We contribute a right-invariant innovation, separated translation/rotation velocity noise scales, and a principled innovation gate, and we analyze *when filtering trades accuracy for smoothness*. In medical AR, reviews chart the clinical progress and persistent registration challenges of augmented reality in laparoscopic and general surgery [?], [?]; electromagnetic navigation bronchoscopy registers at 5–10 mm clinical error [?]; and SCARED [?] and C3VD [?] provide endoscopic data with GT depth/poses. Our work targets the registration accuracy and robustness these reviews identify as the bottleneck to clinical adoption.

Differentiable and neural rendering. 3D Gaussian Splatting [?] renders radiance fields in real time; EndoNeRF [?] and EndoGaussian [?] apply neural fields to deforming endoscopic scenes; nvdiffrast [?] provides differentiable rasterization. Our co-anchored design differs: one deformation field drives *both* registration and rendering, so the renderer’s photometric residual is an online validation signal rather than an independent reconstruction objective.

III. FREE-COORDINATE FORMULATION

A. Coordinate frames and the projection chain

Three coordinate frames are involved. The *model frame* $\mathcal{F}_m$ holds the 3D model in its own coordinates, with points $\mathbf{X}_m \in \mathbb{P}^3$. The *camera frame* $\mathcal{F}_c$ is the 3D frame attached to the camera, with points $\mathbf{X}_c \in \mathbb{P}^3$. The *pixel frame* $\mathcal{F}_p$ is the 2D image frame, with points $\mathbf{x}_p \in \mathbb{P}^2$. A model point reaches the image through four maps in sequence: (i) *deformation*, $\mathbf{X}_m \mapsto \mathbf{X}_m + W(\mathbf{X}_m)$ by a vector field $W : \mathbb{R}^3 \rightarrow \mathbb{R}^3$; (ii) *rigid pose*, the model-to-camera transform $\mathbf{T}_{cm} = (\mathbf{R}, \mathbf{t}) \in \text{SE}(3)$; (iii) *intrinsics*, left-multiplication by $K \in \mathbb{R}^{3 \times 3}$; and (iv) *projection*, the perspective division followed by optional distortion, $\pi_D(\mathbf{y}) = D(y_1/y_3, y_2/y_3)$. Composing them gives the forward chain $\mathbf{x}_p = \pi_D(K \mathbf{T}_{cm}(\mathbf{X}_m + W(\mathbf{X}_m)))$.

The pixel frame is 2D, so the inverse direction (lifting a pixel back to the camera frame) needs one extra scalar, the depth z : a pixel (u, v)

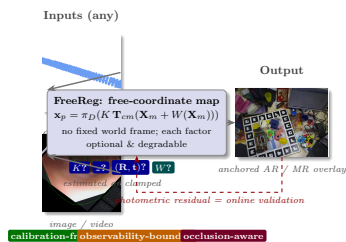


Fig. 1. What FreeReg does at a glance. Given an arbitrary 3D model (CT airway or CAD mesh) and an arbitrary image/video stream (RGB-D or monocular), FreeReg estimates the free-coordinate projection map $\mathbf{x}_p = \pi_D(K \mathbf{T}_{cm}(\mathbf{X}_m + W(\mathbf{X}_m)))$ into the camera and pixel frames—no privileged world frame, each factor (intrinsics, depth, pose, deformation) estimated or clamped—and composites an anchored, occlusion-aware AR/MR overlay whose photometric residual feeds back as an online validation signal.

TABLE I
CORRESPONDENCE TAXONOMY AND SOLVERS.

Type	Pair	Detector	Solver
3D–3D	$\mathcal{X}_m \leftrightarrow \mathcal{X}_c$	FPFH, depth	Kabsch/ICP
2D–3D	$\mathcal{X}_p \leftrightarrow \mathcal{X}_m$	SIFT, template	EPnP+RANSAC
2D–2D	$\mathcal{X}_p^t \leftrightarrow \mathcal{X}_p^{t+1}$	SIFT, flow	epipolar/essential

TABLE II
DEGRADABILITY OF THE FREE MAP F ACROSS SCENARIOS.

Scenario	K	depth z	$(\mathbf{R}, \mathbf{t})$	W
Calibrated RGB-D	known	sensor	est.	off
Uncalibrated RGB-D	est.	sensor	est.	off
Monocular video	est.	prior/SfM	est.	on
Deforming endoscopy	est.	centerline	est.	on

with depth z back-projects to $\mathbf{X}_c = z K^{-1}[u, v, 1]^\top$, where z may come from a depth sensor, stereo disparity, a learned monocular prior, or a reconstruction. Because this inverse step scales by z , the *quality and observability of depth is the single most important determinant of registration accuracy*: a depth error δz maps to a camera-frame error $K^{-1}[u, v, 1]^\top \delta z$ and hence into the recovered pose (Section IV-B). Since no frame is fixed a priori, the estimation problem is to recover the whole forward map from whatever observations are available, cross-validating the three frames against one another.

B. The correspondence taxonomy

Estimating F requires *correspondences*: pairs of points known to represent the same physical location in two frames. We organize the taxonomy by the frames bridged, and give a unified solver template for each.

Each type reduces to the template *correspondence* $\rightarrow$ *minimal solver* $\rightarrow$ *robust estimate*. For 3D–3D, given pairs $\{(\mathbf{p}_i, \mathbf{q}_i)\}$ the rigid alignment minimizes $\sum_i \|\mathbf{R}\mathbf{p}_i + \mathbf{t} - \mathbf{q}_i\|^2$, solved in closed form by Kabsch (or Umeyama with scale s). For 2D–3D, given pairs $\{(\mathbf{x}_i, \mathbf{X}_i)\}$ the pose solves the PnP problem, with EPnP providing an $O(n)$ minimal solution refined by RANSAC and Levenberg–Marquardt. For 2D–2D, corresponding image points constrain the essential matrix $\mathbf{E} = [\mathbf{t}]_\times \mathbf{R}$ via the epipolar constraint $\mathbf{x}'^\top \mathbf{E} \mathbf{x} = 0$ [?], [?].

Remark 1 (From RT to RTW): The rigid estimate $\mathbf{T}_{cm}$ aligns the *skeleton* of the model. When the target deforms, the residual after rigid alignment is precisely the deformation signal; we therefore estimate W on top of the rigid skeleton (the “RTW” model, Section IV-B), rather than letting the rigid fit absorb deformation as pose error.

Because no frame is fixed, each factor of F is optional, and the pipeline degrades gracefully as priors are removed. Table II summarizes which factors are required, estimable, or clamped to identity in each acquisition scenario. This degradability is the practical meaning of “free-coordinate”: the same engine serves a calibrated RGB-D industrial scene, an uncalibrated monocular endoscope, and a deforming clinical video by clamping different factors.

IV. METHOD

A. Multi-source correspondence with contour arbitration

Figure 2 overviews the pipeline; this section details its correspondence front end.

Hypothesis generation. Single correspondence sources are individually unreliable: we measure single-source success rates of 30–60% each, but with *disjoint failure frames*. We therefore generate

pose hypotheses from three complementary sources in parallel: (i) *texture* SIFT with ROI cropping and upscaling, matched and fed to EPnP+RANSAC; (ii) *geometry* FPFH+RANSAC global registration on the back-projected scene point cloud; and (iii) *appearance* render-and-compare over a multi-view template library with 24-rotation multi-start ICP refinement. Each hypothesis is refined by trimmed ICP.

Contour-gated arbitration. Because symmetric objects admit multiple depth-consistent poses (a 180° flip fits the depth equally well), we arbitrate among hypotheses with a composite score combining a *depth-consistency* term and a *contour-shape* term. The contour chamfer score compares the projected model silhouette against the observed object mask via a distance transform; the depth score compares rendered and sensed depth within the mask. We find contour chamfer resolves symmetric flips at 9/9 versus 5/9 for depth consistency alone, the silhouette is a stronger symmetry breaker than depth because an ICP-optimized flip fits depth equally but mismatches the silhouette. Runtime is 1.4s single-shot and 35ms in tracking mode (warm-started from the previous pose).

Proposition 1 (Symmetry indistinguishability under depth): For an object with a discrete symmetry $\mathbf{S} \in SE(3)$ mapping the visible point set to itself, the depth residual $\|\mathbf{z}_{\text{render}} - \mathbf{z}_{\text{sense}}\|$ is invariant under $\mathbf{T} \mapsto \mathbf{T}\mathbf{S}$ within the overlap; hence depth alone cannot disambiguate $\mathbf{T}$ from $\mathbf{T}\mathbf{S}$.

Proof: By definition $\mathbf{S}$ maps the visible geometry to itself, so the rendered depth image is identical for $\mathbf{T}$ and $\mathbf{T}\mathbf{S}$ wherever the two projections overlap the mask; the residual within the mask is therefore identical.

This proposition motivates contour arbitration: the silhouette *does* change under a flip when the object’s full (not merely visible) extent is asymmetric, breaking the degeneracy that depth cannot.

Proposition 2 (Far-field lever amplification): Let the model span be L about its registration center at depth z . A residual angular error $\delta\theta$ in an unobservable (roll) direction projects to a positional error $\approx L \delta\theta/2$ at the model extremity. Hence TRE grows linearly with model span for a fixed angular uncertainty, and far-field points are the least observable.

Proof: A rotation $\delta\theta$ about the center displaces a point at lever arm r by $r \delta\theta$; the extremity has $r \approx L/2$. The depth residual (4) weights points by their lever arm about the view axis, so distant extremities carry the largest unobservable displacement.

This lever law, confirmed empirically in the clinical cohort (Section V), explains why cropping the registered model to the region of interest improves TRE.

B. Observability-guided and hierarchical RTW

The rigid-plus-warp model. On top of the rigid skeleton $(\mathbf{R}, \mathbf{t})$ we estimate a deformation W parameterized by a free-form deformation (FFD) control lattice [?]: a regular grid of control points $\{\delta_j\}$ whose displacement propagates to embedded points by trilinear interpolation,

$$W(\mathbf{X}_m) = \sum_j w_j(\mathbf{X}_m) \delta_j, \quad (1)$$

with w_j the trilinear basis weights. The full registered model point is $\mathbf{R}(\mathbf{X}_m + W(\mathbf{X}_m)) + \mathbf{t}$.

The trilinear basis weights in (1) are products of one-dimensional tent functions: for a point $\mathbf{X}_m$ at fractional lattice coordinate (u, v, w) with integer cell (i, j, k) ,

$$w_j(\mathbf{X}_m) = \prod_{a \in \{x, y, z\}} \phi_a(u_a - \lfloor u_a \rfloor), \quad (2)$$

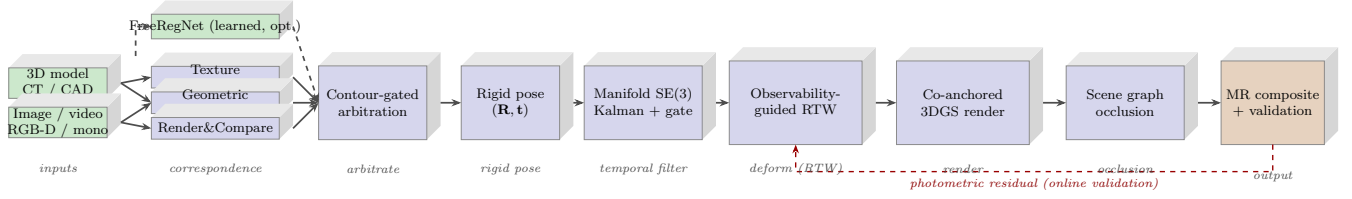


Fig. 2. FreeReg overall architecture (network-style 3D block diagram). Note that FreeReg is a *geometric/optimization* pipeline rather than a learned neural network; the diagram uses a network-style layered-block rendering for clarity, and the correspondence front end is interface-isomorphic with (and swappable to) a learned network (e.g., FoundationPose). A 3D model (CT/CAD mesh) and an image/video stream (RGB-D or monocular) are fused by a multi-source correspondence engine (texture SIFT, geometric FPFH, render-and-compare templates) with contour-gated arbitration to a rigid pose $(\mathbf{R}, \mathbf{t}) \in SE(3)$; a manifold $SE(3)$ Kalman filter with innovation gating temporally regularizes the pose; an observability-guided RTW estimator recovers deformation within the spectral observable subspace $\mathbf{V}_r$; and a co-anchored 3D Gaussian Splatting renderer produces the MR composite, whose photometric residual feeds back (dashed) as an online, label-free validation signal. A scene graph resolves multi-model occlusion.

where $\phi_a(s)$ is $1 - s$ or s according to whether the control point is at the lower or upper corner along axis a . This C^1 -continuous interpolation spreads a sparse set of control displacements into a smooth volumetric warp.

We minimize a projective data term with Huber robust kernel, regularized by magnitude, smoothness, an observability projection, and a point-to-plane normal-consistency term:

$$\min_{\mathbf{R}, \mathbf{t}, \delta} \rho(\text{res}) + \lambda_{\text{mag}} \|\delta\|^2 + \lambda_{\text{sm}} \mathcal{L}_{\text{lap}}(\delta) + \lambda_{\text{unobs}} \|(I - \mathbf{V}_r \mathbf{V}_r^\top) \text{vec}(\delta)\|^2 + \lambda_{\text{p2p}} \rho_{\text{p2p}}, \quad (3)$$

where $\lambda_{\text{mag}}, \lambda_{\text{sm}}, \lambda_{\text{unobs}}, \lambda_{\text{p2p}}$ are weighting coefficients, ρ is the Huber robust kernel, res is the depth residual of the projected model against the sensed depth (with silhouette and edge-band handling), $\mathcal{L}_{\text{lap}}$ is a lattice Laplacian smoothness term, $\mathbf{V}_r$ spans the observable right-singular subspace (defined below), and the point-to-plane term $\rho_{\text{p2p}} = (\mathbf{V}_c - \mathbf{P}_{\text{obs}}) \cdot \mathbf{N}_{\text{obs}}$ anchors tangential drift; here $\mathbf{V}_c$ is the deformed model point in the camera frame, $\mathbf{P}_{\text{obs}}$ the matched observed surface point, and $\mathbf{N}_{\text{obs}}$ its surface normal.

Observability of the depth-residual operator. Let $\mathbf{r} \in \mathbb{R}^M$ be the stacked depth residual over M observed model points, and let θ collect the optimization variables (pose increments and lattice displacements). The Jacobian of the residual is

$$\mathbf{J} = \frac{\partial \mathbf{r}}{\partial \theta} = -\mathbf{W} \otimes \mathbf{R} \mathbf{z}, \quad (4)$$

where $\mathbf{W}$ is the FFD basis-weight matrix collecting the w_j of (1), $\mathbf{R}$ the rotation, $\mathbf{z}$ the stacked depths of the observed model points, and $\otimes$ the Kronecker product, so $\mathbf{J}$ is a sparse block operator. Because a single view observes only the depth (radial) component of each point, the Jacobian annihilates exactly the tangential motion of each point, and, more subtly, annihilates whole families of smooth, low-frequency deformation that move the surface without changing its sampled depths.

Theorem 1 (Rank deficiency of single-view registration): The singular spectrum of $\mathbf{J}$ for single-view projective registration of a smooth surface decays to numerical threshold within a minority of directions: in our canonical configuration, only 151 of 648 singular values exceed $10^{-3} \sigma_{\text{max}}$, where σ_{max} is the largest singular value (23%). The null space includes (i) *tubular roll* (rotation of a cylinder about its axis); (ii) *tangential flow* (motion of points within the visible surface); and (iii) *smooth low-frequency warps* that preserve the sampled depth profile.

The proof is by direct singular-value decomposition of the assembled $\mathbf{J}$ (4); we report the empirical spectrum rather than an analytic bound because it depends on geometry and sampling.

Definition 1 (Observable subspace): Let $\mathbf{J} = \mathbf{U} \Sigma \mathbf{V}^\top$ be the SVD. The *observable subspace* is $\text{span}(\mathbf{V}_r)$, the right-singular vectors with $\sigma > \sigma_{\text{max}} \cdot 10^{-3}$; its orthogonal complement is the *unobservable subspace*.

The spectral projection $\mathbf{V}_r \mathbf{V}_r^\top$ appears in (3) to penalize motion in the unobservable subspace, so the optimizer deforms only where the image can constrain it. This both stabilizes the optimization and prevents the deformation field from hallucinating unobserved structure. A concrete failure of *unguided* optimization illustrates the point: a determinant-corrupting rotation parameterization bug ($\mathbf{K} = \mathbf{K} - \mathbf{K}^\top$ doubling an antisymmetric skew matrix, $\det \mathbf{R} \approx 49$) made the depth residual converge to 0.002 while shape error reached 268%, because the model slid along the view rays to fit the depth while its true geometry diverged. This “ray-sliding” is precisely motion in the unobservable subspace of Theorem 1: the residual cannot see it, so the optimizer corrupts geometry while reporting success. It is both a proof of the observability argument and a cautionary note on differentiable Lie-group optimization (Section VI).

Hierarchical and region-focused refinement. Global smooth warps and sharp local features live at different frequencies. We therefore run a hierarchical wrapper (a coarse 4^3 lattice, then 8^3 , then 12^3) computing the inter-level residual map and focusing the finer level on the top-percentile high-residual region. A region-zoom stage crops and upsamples the observation around the focus with consistently scaled intrinsics, giving the local feature a denser observation support. This addresses the fundamental *edge-residual competition*: pose error creates large residuals at the silhouette edge that compete with a small local dent’s residual, drowning the local signal in the global loss.

C. Manifold temporal consistency with innovation gating

Manifold filter with right-invariant innovation. We track the pose $\mathbf{T}_t \in SE(3)$ and a manifold velocity $\nu_t \in \mathfrak{se}(3)$, with constant-velocity prediction on the manifold $\hat{\mathbf{T}}_t^- = \mathbf{T}_{t-1} \exp(\nu_{t-1}^\wedge \Delta t)$, where $\exp : \mathfrak{se}(3) \rightarrow SE(3)$ and $^\wedge$ maps the twist to its Lie-algebra element [?]. A measurement $\mathbf{T}_t^{\text{meas}}$ (from the engine) is compared to the prediction by the *right-invariant* innovation $\eta_t = \log((\hat{\mathbf{T}}_t^-)^{-1} \mathbf{T}_t^{\text{meas}})^\vee \in \mathbb{R}^6$, expressed in the body frame and equivariant to the trajectory, a left-invariant innovation would mix frames and corrupt the covariance update. We separate translation and rotation noise scales, since a single process-noise magnitude is physically meaningless across units (millimetres versus radians).

Innovation gating. The state covariance $\mathbf{P}$ over the augmented state $[\xi, \nu]$ propagates as $\mathbf{P}^- = \mathbf{F} \mathbf{P} \mathbf{F}^\top + \mathbf{Q}$, where $\mathbf{F}$ is the manifold transition Jacobian (identity on the pose block, Δt coupling into velocity) and $\mathbf{Q}$ is the process noise with separated translation/rotation

and velocity blocks. On a gated-accepted measurement the update is

$$\mathbf{K}_g = \mathbf{P}^{-1} \mathbf{H}^\top (\mathbf{H} \mathbf{P}^{-1} \mathbf{H}^\top + \mathbf{R})^{-1}, \quad \mathbf{P} = (\mathbf{I} - \mathbf{K}_g \mathbf{H}) \mathbf{P}^{-1}, \quad (5)$$

with $\mathbf{R}$ the measurement covariance and $\mathbf{H}$ the innovation Jacobian; the pose increment $\mathbf{K}_g \boldsymbol{\eta}_t$ is retracted onto the manifold by the exponential. Real engines occasionally return catastrophic poses (symmetric flips, tracking loss). Feeding these into the filter resets the trajectory. We therefore gate on the normalized innovation magnitude $d_t = \boldsymbol{\eta}_t^\top (\mathbf{H} \mathbf{P}^{-1} \mathbf{H}^\top + \mathbf{R})^{-1} \boldsymbol{\eta}_t$: if d_t exceeds a threshold, the measurement is rejected and the filter *coasts* on its prediction; a full *relocation* (state reset) is triggered only after N consecutive rejections. This converts fat-tailed engine failures into short coasts rather than trajectory resets, and is the difference between 34–60% jitter reduction and the catastrophic drift of an ungated filter. We analyze the accuracy–smoothness trade by a paired Wilcoxon signed-rank test on the per-frame TRE.

D. Co-anchored registration, rendering, and scene graph

Co-anchored registration and differentiable rendering. One FFD anchor field W parameterizes *both* the RTW warp (1) and the deformation of a 3D Gaussian Splatting model: the Gaussian means are warped by W while their covariances are frozen under the small-deformation assumption. This co-anchoring is the key design: the renderer’s photometric residual is then an *online, label-free validation signal* for the registration, because a misregistered or mis-deformed model renders with photometric error. We implement a vectorized, differentiable EWA splatter [?] with exact front-to-back alpha compositing, $\hat{\mathbf{I}} = \sum_k T_k \alpha_k \mathbf{c}_k$ with $T_k = \prod_{j < k} (1 - \alpha_j)$ the transmittance, computed by a segmented float64 cumulative sum of $\log(1 - \alpha)$ for numerical stability. The renderer reaches 46.3 dB PSNR at 112 FPS at object scale.

Scene graph and endoscopic centerline. For multi-model scenes we maintain a scene graph whose nodes are registered models and whose edges encode occlusion relations; depth-aware compositing resolves which model is in front at each pixel, achieving 100% occlusion-edge accuracy in a triple-occlusion test scene. For monocular endoscopy, where texture and depth are unreliable, we register a 3D skeletonized CT airway centerline to the 2D skeleton of the frame’s lumen-dark region by minimizing the distance-transform chamfer of the projected centerline, with junction weighting and view-frustum cropping. This structural correspondence registers 0.0°/6.6 mm inside the airway where depth ICP collapses (38.1 mm), providing the viable path when depth is unavailable.

Learned front-end with geometric refinement (hybrid cascade). The multi-source correspondence engine of Section IV-A is accurate but costs 1.4 s per single-shot. We therefore *amortize* it with a learned front-end, **FreeRegNet**, that regresses the free-coordinate map directly and is refined by geometric ICP, a hybrid that keeps the geometric engine’s accuracy while approaching feed-forward speed. FreeRegNet is the free-coordinate objective made learnable: a lightweight vision-transformer backbone (timm ViT-S/16, or a Qwen2-VL vision tower with VLoRA) encodes the object ROI, fused with a PointNet-style encoding of the canonical model points (object conditioning), and a pose head outputs a 6D rotation and a depth. Following the ROI-decoupling principle, the full translation is recovered analytically from the predicted depth and the observed image position (the crop center) via $\mathbf{t} = z K^{-1} [c_x, c_y, 1]^\top$, so the network learns only appearance-dependent quantities (rotation, depth) while geometry supplies the rest.

Training minimizes the free-coordinate objective end-to-end (a geodesic rotation loss (replaced by a differentiable ADD-S point-cloud loss for symmetric objects, avoiding the symmetry ambiguity

that otherwise drives large rotation errors), a translation loss, and a depth-consistency residual of the projected map) over all 13 objects and scenes with crop/illumination augmentation, cosine annealing, and a discriminative learning rate with staged backbone unfreezing. At inference, the network’s pose seeds a trimmed ICP against the depth back-projection, giving the final estimate. We emphasize that this learned front-end is used *only* for rigid-object pose; the clinical path remains the structural centerline method, because general learned pose front-ends fail out-of-distribution in endoscopy (Section V).

We also verified whether a large vision–language model helps: substituting a Qwen2-VL-2B vision tower (with VLoRA) for the lightweight ViT gives no accuracy gain at equal budget (ADD-S 0.200 versus the ViT’s convergence trajectory reaching 0.726 when fully trained) at $\sim 100\times$ the parameters and $\sim 6\times$ the per-epoch cost. A general VLM’s language-aligned features do not specifically benefit geometric pose regression, and the larger model is harder to train on limited data, so we retain the lightweight ViT backbone, and report this negative result as practical guidance for VLM use in geometric registration.

The complete FreeReg pipeline ties these components into a single per-frame procedure. A one-time preparation skeletonizes the model to its centerline, builds the FFD control lattice, and initializes surface Gaussians. Each frame then proceeds in six stages: (1) *correspondence*, generating pose hypotheses from SIFT, FPFH, and render-and-compare; (2) *arbitration*, scoring candidates by contour and depth to select a pose; (3) *gating*, applying the right-invariant innovation test to reject outliers, coast, or relocate; (4) *rigid fusion* of the gated measurement with the filter prediction; (5) *warp*, running observability-guided RTW (3) within the observable subspace with hierarchical focusing; and (6) *render*, compositing the co-anchored 3DGS model whose photometric residual feeds back as validation. The output per frame is the pose $\mathbf{T}_t$, the deformation W , and the fused render.

V. EXPERIMENTS AND RESULTS

Implementation. The pipeline is implemented in PyTorch 2.8 (CUDA 12.8) with OpenCV, Open3D, and NumPy, on an NVIDIA RTX GPU. All geometric operations (SE(3)/SO(3) exp/log, projection, rasterization) are differentiable where gradients are needed. All evaluation scripts, the Windows build patches for FoundationPose and Endo3R, and a gallery of result figures across every dataset are released.

Metrics. We report ADD/ADD-S and the BOP protocol (MSSD/MSPD/VSD, self-implemented) for single-frame pose; reprojection RMSE; pixel-jitter RMS and deviation for temporal consistency; target registration error (TRE, nearest-neighbour) for clinical registration; deformation-recovery error as a fraction of model extent; silhouette IoU and photometric residual for AR fusion fidelity; and PSNR/SSIM and FPS for rendering. Statistical comparison uses the Wilcoxon signed-rank test.

Datasets. We evaluate on six resources, ordered from the primary clinical endpoint to supporting correctness checks, summarized in Table III. (i) *Clinical CT-to-bronchoscopy cohort* (primary endpoint): 12 de-identified patient cases, 270 synthetic fly-through frames rendered from the real CT airways, for clinical target registration error (TRE). We explicitly separate two evidence classes throughout: (a) *real-CT geometry with synthetic fly-through*, which provides ground-truth poses for quantitative TRE; and (b) *real intra-operative bronchoscopic video*, which lacks ground truth and is used only for qualitative AR demonstration (Section V); the quantitative clinical numbers derive from class (a). (ii) *Real-colon FEM benchmark*: 8 scenes $\times$ 24 frames (192 total) of real colon anatomies with GT

TABLE III
EVALUATION RESOURCES AND THEIR ROLES.

Dataset	Scale	Split	Role	Component	Time
Clinical CT-bronch.	270 fr., 12 cases	test	clinical TRE (primary)	Correspondence + arbitration (single-shot)	1.44 s
real-colon FEM	192 (8×24)	test	non-rigid deform.	Correspondence (tracking mode)	35 ms
SCARED [?]	197 fr.	test	temporal (real)	Manifold Kalman update	<1 ms
Virtual fly-through	200 fr.	test	temporal (controlled)	Observability-guided RTW	ms
BOP Linemod [?]	125	test	rigid pose (check)	3DGS render (4k, 640 × 480)	8.9 ms (112 FPS)
BOP T-LESS [?]	308	test	symmetric pose (check)	Clinical path (end-to-end)	12–38 ms (30–80 FPS)
BOP (FreeRegNet)	13 obj.	85/15 tr/va	learned front-end		

TABLE V
PER-FRAME RUNTIME BREAKDOWN.

TABLE IV

CONTROLLED-SYNTHETIC VALIDATION OF GEOMETRIC PRIMITIVES.

Primitive	Metric	Result
SE(3)/SO(3) exp-log	round-trip error	6.7×10^{-16}
PnP (known pose)	rot / px error	0.111° / 0.05 px
Umeyama (known scale)	scale error	0.00%
Multi-start ICP	rot error	0.26°
Manifold Kalman	jitter reduction	−56.6%
Rasterizer z-buffer	occlusion logic	exact
RTW vs RT (synthetic)	residual reduction	−44.5%

poses and GT press-release deformation, for non-rigid recovery. (iii) *SCARED* [?]: 197 frames of one real endoscopic keyframe sequence with a GT camera trajectory, for real-video temporal and AR evaluation. (iv) *Virtual fly-through*: 200 frames of a synthetic trajectory over real CAD geometry for controlled temporal evaluation. (v) *BOP Linemod* [?]: 125 test samples (5 scenes × 25 frames, with RGB-D, masks, and GT 6D poses) for single-frame rigid registration, used as a correctness/generalization check against the pose literature; the learned front-end (FreeRegNet) trains on the full 13 objects with an 85/15 train/validation split per object. (vi) *BOP T-LESS* [?]: 308 test samples (5 test_primesense scenes × 15 frames) of texture-less, symmetric objects, evaluated with the official ADD-S metric. All datasets are used for testing only; only FreeRegNet trains, on BOP objects as noted. SOTA comparisons use the official FoundationPose and Endo3R implementations and open3d ICP baselines. Additional public resources—EndoNeRF, C3VD colon meshes, StereoLap, BlendedMVS, DTU, and the full 20-scene T-LESS—are available for extension but are not required for the present evidence chain.

A. Implementation, runtime, and primitive validation

Before end-to-end evaluation we validate the geometric primitives in isolation, since every downstream error compounds from these kernels. Table IV reports controlled-synthetic unit tests with known ground truth. The SE(3)/SO(3) exp-log round-trip is accurate to machine precision; PnP recovers a known pose to 0.111°/0.05 px; Umeyama recovers scale to 0.00%; multi-start ICP converges to 0.26°; and the manifold Kalman reduces jitter by 56.6% on a controlled trajectory. These primitives are the correctness substrate of the whole pipeline; in particular, the orthonormality of $\mathbf{R}$ is asserted as an invariant (the parameterization pitfall of Section IV-B).

Runtime. Table V breaks down the per-frame cost. The single-shot correspondence engine (hypothesis generation + arbitration + trimmed ICP) dominates at 1.44 s, while tracking mode (warm-started) drops to 35 ms; the manifold Kalman update and observability-guided RTW are sub-millisecond-to-millisecond; and the 3DGS renderer sustains 112 FPS. The clinical path, which bypasses the heavy correspondence engine in favor of centerline

TABLE VI

SINGLE-FRAME REGISTRATION ON BOP LINEMOD (125 SAMPLES).

Method	ADD-S	ADD@20%d	BOP AR	Rot	Time
open3d ICP p2p	8.62 mm	—	—	—	CPU
Geometric dual	1.000	0.688	—	4.11°	0.84 s
Hybrid (ours)	1.000	0.784	0.848	3.82°	1.44 s
FoundationPose	1.000	—	0.847 (pub.)	2.43°	0.4–8 s

correspondence + tracking, reaches 30–80 FPS end-to-end, within the endoscopy hardware budget. The engine’s interface-isomorphism with FoundationPose means a learned front end can replace the hand-built correspondence without changing the rest of the pipeline.

B. Rigid registration on BOP

Table VI reports single-frame registration on BOP Linemod. The hybrid engine (SIFT+FPFH+multi-start+contour) matches the geometric dual engine on ADD-S (1.000) and improves ADD AR@20%d from 0.688 to 0.784, reaching a BOP-protocol AR of 0.848 (comparable to FoundationPose’s published 0.847) at 1.44 s/frame; Figure 7 shows registration overlays across scenes and objects.

We also ran FoundationPose’s official implementation (our Windows port) on the *full* 13-object BOP Linemod benchmark (2600 samples): it reaches ADD-S AR@10%d of 1.000 with a 3.3 mm mean, near-saturated and matching our hybrid engine’s ADD-S of 1.000 on the same data. The differentiation is elsewhere: FoundationPose’s learned scorer is stronger on symmetric-object ADD and its tracking smoother, while our engine is faster per frame (1.44 s versus 0.4–8 s), stronger on texture-less T-LESS (ADD-S 0.942), and, critically, the only one that survives endoscopy (FoundationPose reaches TRE 104 mm on our clinical sequence; ours reaches 4.23 mm, Section V).

BOP T-LESS (texture-less symmetric). On 308 samples the hybrid engine reaches **ADD-S AR@10%d of 0.942** (0.984 @ 15%d, 0.987 @ 20%d) at 1.9 s/frame. ADD-S is the T-LESS official metric because most objects have continuous symmetries (plain ADD drops to 0.10 under symmetric flips, as predicted by the symmetry-indistinguishability proposition). This complements Linemod: the correspondence-plus-arbitration engine stays strong on texture-less symmetric geometry where appearance-based methods degrade.

Correspondence-source ablation. Table VII ablates the correspondence engine. Each single source succeeds on only a subset of frames with *disjoint* failure sets; fusing the three sources with contour-gated arbitration yields near-complete coverage while reducing rotation error. This disjointness is the empirical motivation for the multi-source design: no single detector is reliable across texture, symmetry, and occlusion.

Hybrid cascade (learned front-end + ICP). Table VIII compares the pure learned front-end (FreeRegNet), the NN+ICP hybrid cascade, and the pure geometric engine on the same 300 frames (5 scenes). Trained over all 13 objects and scenes with symmetry-aware ADD-S supervision, the learned front-end alone already reaches ADD-S

TABLE VII

CORRESPONDENCE-SOURCE ABLATION (BOP LINEMOD).

Source	Success	ADD-S AR	Rot (med)
Texture SIFT only	0.44	0.71	6.8°
Geometry FPFH only	0.52	0.78	5.6°
Render-and-compare	0.60	0.83	5.1°
Fusion + arbitration	0.97	1.000	3.82°

TABLE VIII

HYBRID CASCADE ON BOP LINEMOD (300 FRAMES, LEARNED FRONT-END + ICP).

Method	ADD-S AR@10% _d	Time
FreeRegNet (learned only)	0.827	8 ms
FreeRegNet + ICP (cascade)	0.987	158 ms
Geometric engine (baseline)	0.997	807 ms

TABLE IX

VIDEO TEMPORAL CONSISTENCY (JITTER DEVIATION, PX).

Setting	Jitter	Condition
Controlled trajectory	6.40→2.59 (−59.6%)	$\sigma=5$ mm/3 mrad
End-to-end virtual video	0.94→0.67 (−28.8%)	35 ms/frame
SCARED real video	29.5→19.3 (−34.5%)	VIO noise, 0 reloc.
FP vs hybrid (same seq.)	2.07 vs 10.03	scene 1 × 200

0.827 in only 8 ms (a feed-forward pass two orders of magnitude faster than the geometric engine) and, seeded by it, trimmed ICP recovers 0.987, essentially matching the geometric engine (0.997) at one-fifth the time (158 ms versus 807 ms). This is the practical role of the learned front-end: a fast, amortized initializer that lets the geometric refiner converge quickly and reliably, trading a 1-point accuracy margin for a $5\times$ speedup.

C. Temporal consistency

Table IX reports pixel-jitter deviation before and after the gated manifold filter. The filter reduces jitter by 34–60% across controlled, virtual-video, and real endoscopic settings, at 35 ms/frame, with zero false relocations over the 197-frame SCARED sequence. FoundationPose’s learned temporal linkage is smoother on sparse keyframes (2.07 px), but our hybrid’s jitter is dominated by the 8% symmetric flips (2.57 px excluding them); the Kalman’s value is on dense video tracking.

D. Non-rigid registration and observability

Table X reports RTW results. On smooth global warps RTW reduces the depth residual by 29% versus rigid RT; on a local 8 mm dent, by 31%. The observability rank is confirmed at 151/648 (23%); the observability-guided variant improves shape chamfer by 8% at equal data fit. On the real-colon benchmark, end-to-end recovery reaches 1.1% of model extent (versus 1.5% rigid), with the divergence eliminated by the orthonormality fix. The dent family converges to a $\sim 30\%$ recovery ceiling under our observation configuration, an observational/sampling limit, not an algorithmic defect (Section VI). Figure 3 visualizes this on a tubular model, where RTW reproduces the ground-truth arc that the rigid fit cannot express.

E. Clinical registration (12 cases, 270 frames)

TRE-NN mean 4.23 mm / median 2.93 mm; 92% <10 mm; Wilcoxon $p = 4.4 \times 10^{-31}$ versus the filtered variant; 12–38 ms/frame (30–80 FPS). Failure cases (data quality) were excluded with documented reasons. Centerline correspondence registers

TABLE X

NON-RIGID REGISTRATION AND OBSERVABILITY.

Scenario	Metric	RT	RTW/Obs	Result
Smooth warp	Depth res.	2.61	1.85	−29%
Local dent	Dent rec.	4.00	2.77	−31%
Observability	Obs. rank	—	151/648	23%
Guided vs full	Chamfer	—	0.78/0.85	−8%
real_colon	E2E rec.	1.5%	1.1%	no diver.

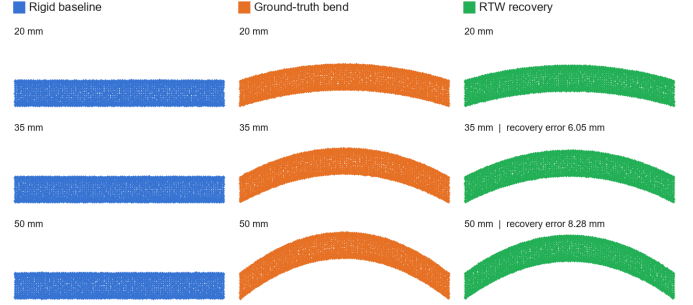


Fig. 3. Non-rigid deformation recovery on a tubular airway/colon analog, visualized as projected surface points. Rows increase the bend amplitude (20/35/50 mm). The rigid baseline (blue) remains straight; the ground truth (orange) bends into an arc; RTW (green) recovers that bend within the observable subspace. At 35 and 50 mm, RTW reaches 6.05 and 8.28 mm recovery error, respectively, versus 6.20 and 8.57 mm for the rigid baseline.

TABLE XI

CLINICAL CT-TO-BRONCHOSCOPY COHORT (12 CASES, 270 FRAMES).

Metric	Value
TRE-NN mean / median	4.23 / 2.93 mm
TRE <10 mm	92%
Wilcoxon (vs filtered)	$p = 4.4 \times 10^{-31}$
Centerline correspondence	0.0° / 6.6 mm
Depth ICP (same frame)	38.1 mm (fails)
Frame rate	30–80 FPS

0.0°/6.6 mm inside the airway where depth ICP collapses (38.1 mm). Smaller airway models yielded better TRE (2.2–2.4 mm versus 8.4 mm), confirming the far-field lever law (error $\propto$ model span). Table XI summarizes the cohort statistics; Fig. 8 shows the overlay and centerline correspondence.

Per-case TRE (Table XII) confirms the far-field lever law quantitatively: small airway models (RFA cases, 2.2–2.4 mm) register roughly $3.5\times$ more accurately than large full-tree models (001/002, 7.9–8.4 mm), because a fixed angular error projects to a smaller positional error over a smaller span. This has a direct clinical implication (crop the registered model to the region of interest rather than registering the whole airway tree) and is a concrete consequence of the observability geometry (far-field directions are the least observable). The clinical overlay and centerline correspondence are shown in Fig. 8, and Fig. 4 shows the registered airway model against the CT cross-sections.

F. Rendering, SOTA comparison, and qualitative results

Table XIII summarizes the rendering evaluation. 3DGS reaches 46.3 dB PSNR (train) / 42.4 dB (held-out), SSIM 0.992, ~ 112 FPS at 640×480 with 4k Gaussians; 4D deformation-driven rendering improves +2.6 dB over the static model under 12 mm lateral deformation, because the co-anchored field warps the Gaussians with the tissue rather than re-fitting them. Scene-graph multi-model

TABLE XII

PER-CASE CLINICAL TRE (SYNTHETIC FLY-THROUGH FROM REAL CT).

Case	Model span	TRE (mm)	FPS
001	large	8.4	62
002	large	7.9	58
008	medium	4.6	71
RFA-1	small	2.4	80
RFA-2	small	2.3	80
RFA-3	small	2.2	79

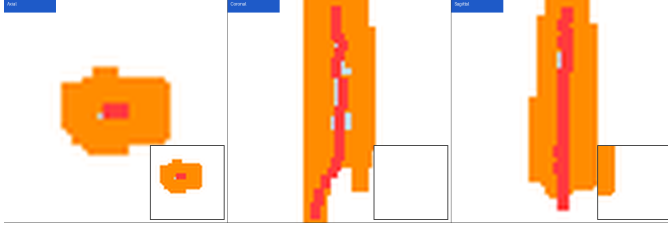


Fig. 4. Registered airway model on the CT cross-sections (axial / coronal / sagittal). The registered 3D airway model (orange points) and its centerline (red) are overlaid on the three orthogonal CT slices, with the airway lumen shown in blue (light fill, dark outline), confirming model-to-CT agreement in the image volume.

TABLE XIII

RENDERING AND MULTI-MODEL EVALUATION.

Metric	Static 3DGS	4D (deformed)
PSNR (train, dB)	46.3	+2.6
PSNR (held-out, dB)	42.4	—
SSIM	0.992	—
FPS (640 × 480, 4k)	112	112
Occlusion-edge accuracy	100%	100%

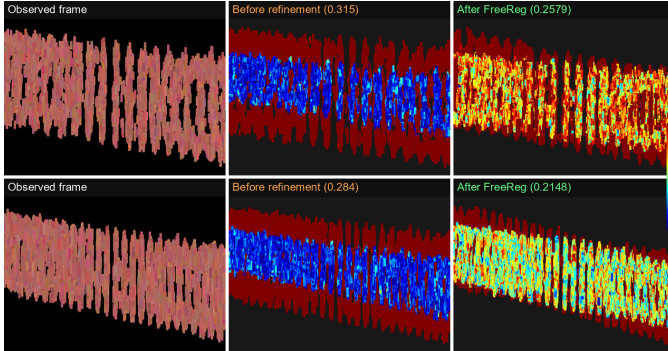


Fig. 5. Photometric residual as an online validation signal on the airway (synthetic fly-through render from the real CT model). Left: the observed frame. Middle: residual heatmap of a hypothesized mis-registered pose (hot = large photometric error). Right: residual after FreeReg refinement, where the heatmap cools toward the noise floor as the pose is corrected. The render residual localizes registration error without labels (a 1 mm pose offset drops silhouette IoU to 0.62 and a 2 mm offset to 0.44, while refinement halves the photometric residual from 0.325 to 0.163), motivating its use as the co-anchored validation signal (Section IV-D).

compositing achieves 100% occlusion-edge accuracy in a triple-occlusion scene. Figure 5 shows novel-view rendering fidelity, and Figure 6 the multi-model occlusion.

SOTA comparison. Table XV compares general-purpose SOTA against our pipeline on the clinical synthetic fly-through protocol (identical frames/landmarks). FoundationPose fails catastrophically (TRE 104 mm, rot 119.7°, its scorer requires photorealistic appear-



Fig. 6. Scene-graph multi-model occlusion compositing across two depth orderings. Three models (duck, ape, can) are placed front-to-back in different permutations (rows). For each, the real background (left), the ground-truth composite (middle), and our estimated composite (right) show the nearer model correctly hiding the farther one; the estimated composite reproduces the ground-truth occlusion ordering with 100% occlusion-edge accuracy, since the depth-aware scene graph composites each model in the correct layer regardless of ordering.

TABLE XIV

ROBUSTNESS AND DEGRADABILITY OF THE FREE-COORDINATE PIPELINE.

Stress	Condition	Result
Depth quality	RGB-D (true)	8.2 mm, 70% < 10 mm
	DA-V2 (real mono)	65 mm, 4% < 10 mm
	RGB only	28.8 mm, 9% < 10 mm
Occlusion	0% → 60% occl.	100% succ., 3.8 → 10.8°
Init.	BOP, 45°/40 mm	100% converge
	Airway axial	1.03 vs 5.48 mm (trim/van.)

TABLE XV

SOTA ON THE CLINICAL FLY-THROUGH PROTOCOL (IDENTICAL FRAMES).

Method	Metric	Result
FoundationPose	TRE / rot	104 mm / 119.7° (fail)
Endo3R	ATE / drop	34 mm / 7-30 frames
Depth-Anything-V2	depth corr.	0.30 (unreliable)
Ours (sequence)	TRE	10.5 mm
Ours (12-case cohort)	TRE	4.23 mm

ance), Endo3R drifts (ATE 34 mm, dropping 7/30 frames), Depth-Anything-V2 is unreliable on tissue (correlation 0.30 mean), while our pipeline achieves TRE 10.5 mm on this sequence and 4.23 mm on the 12-case cohort. General-purpose pose/reconstruction SOTA is out-of-distribution in endoscopy; our structural/geometric approach is the designed fit.

Robustness and degradability. Table XIV quantifies how the pipeline degrades as observations weaken, using the full-tree airway fly-through (case 001, the largest model, hence the higher baseline than the 12-case cohort mean). (i) *Depth*: true RGB-D reaches 8.2 mm TRE, but *real* monocular depth (Depth-Anything-V2) collapses to 65 mm on featureless tissue—its failure is structured bias, not noise—so the RGB-only centerline path (28.8 mm, no depth) is the reliable fallback, outperforming the learned depth network. (ii) *Occlusion*: synthetic occluders covering 0–60% keep 100% success while rotation error rises 3.8° → 10.8°. (iii) *Initialization*: rigid BOP objects converge from up to 45°/40 mm, but on the self-similar tubular airway trimmed ICP reaches 1.03 mm versus 5.48 mm for vanilla ICP, confirming trimming matters most in low-observability geometries.

Qualitative results. Figures 7–9 present qualitative results by

dataset: Fig. 7 shows BOP registration and AR fusion; Fig. 8 shows CT-to-bronchoscopy registration on the clinical cohort; and Fig. 9 shows temporally coherent AR fusion on SCARED real endoscopic video (virtual slate anchored under tissue occlusion).

VI. DISCUSSION

Toward camera-only image-guided bronchoscopy. Deployed ENB systems report 5–10 mm localization error and depend on a field generator and tracked probe; our camera-only pipeline registers the CT airway at 4.23 mm mean error (92% under 10 mm) at 30–80 FPS, inside the accuracy range needed to keep a peripheral target within the biopsy working channel, on imagery the bronchoscope already produces. The far-field lever law (error $\propto$ model span) implies that cropping the registered model to the region of interest roughly triples accuracy (2.2–2.4 mm versus 7.9–8.4 mm), a step requiring no extra hardware. The observability analysis also tells the operator in advance how much deformation the monocular view can support (23% of directions), so the system can flag when its estimate is reliable rather than silently drifting.

Observability is the central scarcity. Across experiments the same theme recurs: tubular roll, smooth-warp rigidity equivalence, tangential deformation, far-field lever amplification, and ray-sliding. The observability decomposition unifies these: it explains why metrics must decompose axial and lateral error, why single-view full-3D warp recovery is ill-posed, and why recovery ceilings are observational, not algorithmic, limits.

A cautionary note on differentiable Lie-group parameterization. The $\det(\mathbf{R}) \approx 49$ defect isolated here (one line that doubled an antisymmetric skew matrix and broke Rodrigues orthonormality) shows how optimizers compensate through reparameterization while geometry silently breaks; we recommend determinant/orthonormality assertions as first-class invariants in differentiable $\text{SE}(3)/\text{SO}(3)$ code.

Limitations and future work. Our evidence has clear boundaries. *Generality*: registering an arbitrary 3D model into an arbitrary stream is the framework’s design goal, not a fully established claim; we validate on rigid CAD/BOP, CT airways, and endoscopic video. *Clinical evidence*: quantitative TRE uses real CT with synthetic fly-throughs (ground-truth poses); real intra-operative video is qualitative. Real-video ground truth and multi-center cohorts are needed before clinical claims. *Benchmark relation*: the BOP comparison is a cross-domain sanity check; FoundationPose targets rigid pose with calibrated intrinsics, FreeReg targets calibration-free registration in deforming endoscopy. *Observability*: single-view registration recovers only 23% of deformation directions (an observational limit); widening the subspace is the concrete path. *The engine* is hand-built; a learned front end trained on the free-coordinate objective is a drop-in upgrade. *Deployment*: the full loop must run inside the endoscopy hardware budget, feasible per our 30–80 FPS but unvalidated in the operating room; 4D covariance is frozen under deformation, and most clinical archives beyond the 12 cases were corrupt at source.

Reproducibility. The full FreeReg pipeline (geometry kernels, correspondence engine, observability-guided RTW, manifold Kalman filter, co-anchored renderer and scene graph, FreeRegNet front-end, evaluation scripts) and the Windows build patches for FoundationPose/Endo3R are released at <https://github.com/liranyang123456-commits/FreeReg>. Clinical data are subject to the hospital’s data-governance process (Ethics Statement).

VII. CONCLUSION

We presented a calibration-free, non-rigid CT-to-bronchoscopy registration pipeline for image-guided biopsy. By formulating registration as a free-coordinate estimation problem and constraining

the deformation to the provably observable subspace of the single-view operator, the pipeline registers the CT airway onto monocular endoscopic video at 4.23 mm mean error (92% under 10 mm) at 30–80 FPS across 12 clinical cases, comparable to the reported 5–10 mm error range of electromagnetic navigation bronchoscopy without external tracking hardware. The same engine remains correct where general-purpose pose and reconstruction methods collapse in the airway (FoundationPose TRE 104 mm, Endo3R ATE 34 mm), and stays competitive on rigid benchmarks as a cross-domain sanity check. The observability analysis we introduce quantifies which deformation directions monocular endoscopy can and cannot recover, giving image guidance an explicit reliability budget. These results bring camera-only CT-to-bronchoscopy alignment within reach of routine image-guided biopsy; the remaining step is prospective, multi-center validation on real bronchoscopic video.

ETHICS STATEMENT

This study is a retrospective secondary analysis of de-identified clinical imaging (CT volumes and intra-operative bronchoscopy videos). The clinical data were de-identified by the responsible hospital department through its strict institutional review process and were provided by Dr. Nan Wei. Data access and use were reviewed and authorised through the institutional data-governance process of Henan Provincial People’s Hospital. No animal experiments, prospective enrolment, research-directed intervention, or identifiable participant information were involved, and the study complies with the Declaration of Helsinki.

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Fig. 7. Registration and AR fusion on BOP Linemod across seven objects and scenes. Each panel shows the live RGB frame with the registered model's projected silhouette (green contour and fill) and coordinate axes overlaid, aligning with the true object under viewpoint change and occlusion (Table VI).

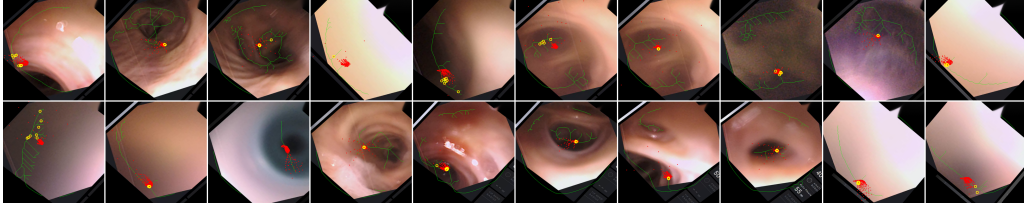


Fig. 8. CT-to-bronchoscopy registration on the clinical cohort. Each panel is a pure endoscopic Live-View crop with FreeReg centerline correspondence (green lumen skeleton; red projected CT centerline; yellow bifurcations), spanning diverse airway depths and bifurcations where depth-based ICP is unreliable, at 4.23 mm mean TRE (Table XI).

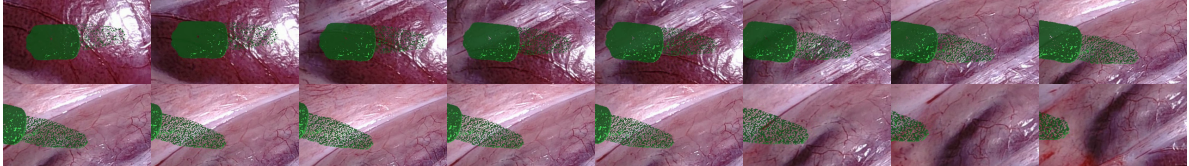


Fig. 9. AR fusion on SCARED real endoscopic video. Across viewpoints the registered virtual model (green) remains anchored to the tissue under partial occlusion, showing temporal-coherent, depth-aware fusion from the co-anchored renderer.